Application of MobileNet-v1 for Potato Plant Disease Detection Using Transfer Learning

Anshuman Singh*
Amity School of Engineering and Technology, Amity University, Lucknow Campus, Lucknow, Uttar Pradesh, India

Sumita Mishra*
Amity School of Engineering and Technology, Amity University, Lucknow Campus, Lucknow, Uttar Pradesh, India
mishra.sumita@gmail.com

Vineet Singh
Amity School of Engineering and Technology, Amity University, Lucknow Campus, Lucknow, Uttar Pradesh, India

ABSTRACT

Infectious diseases have troubled farmers continuously by spreading throughout crops. Thus, a proper identification of such disease is obligatory for the timely treatment which can save the money and efforts of small-scale farmers. The recent advancement in deep learning has provided a way to contribute to the sector of agriculture. In this paper deep learning based MobileNet architecture is employed to identify potato plant lesion characteristic. The application of transfer learning is accomplished by freezing the base layers and training only top 23 layers containing the added classifier layer. The model is then trained further to improve performance. The frozen layer weights of this pretrained model remained constant during training while the top layer weights are constrained by fine tuning to quit generalize feature map and get associated with specific features of new dataset. This enhances the model performances and gives 99.83 % accuracy in the image classification on the leaves of potato plant into the categories of infected disease. The experimental results demonstrate the feasibility of this procedure on portable devices.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Machine learning approaches; Neural networks; • **Information systems** → Information systems applications; Mobile information processing systems.

KEYWORDS

Convolutional neural network, MobileNet, Depth wise separable convolution, Transfer learning

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1 INTRODUCTION

In the sector of agriculture India stands second in Potato production. It has shown tremendous increase in cultivation of potato since last fifteen years, making the knowledge of the Potato disease as prerequisite for keep production intact therefore the effective management of potato disease such as Early and Late blight depend not only on farmer's knowledge but also integrated management methods in their daily practices [1].

The impact of climatic change due to global warming has led to the evolution of new disease and extension in the period of existing diseases among crops. The problematic core highlighted by this paper is late blight and early blight disease in potato plant. Late blight-It is one of the most destructive disease wherever potato is grown. The other disease is Early blight which is the principal foliar disease of potato worldwide. The basic vandalization by early blight leads to untimely ruination of the potato plants, resulting in tuber yield reduction. The yield loss due to leaf vandalized by early blight in potato vary by location, season of cropping, cultivar, and the stage of potato maturity. The lower leaves of Potato are the place where first Early Blight symptoms starts, and they become chlorotic and abscise prematurely. Current approach towards identification and management of these diseases involves manual inspection by plant disease experts from different government agriculture research organizations to advise the farmers. Nevertheless, all the farmers do not have access to experts. Thus, there is need to develop quick, easily accessible, and inexpensive tools for automatic identification of plant diseases. Deep learning-based computer vision technology is an attractive approach for automatic crop disease identification.

Considering the importance of deep convolutional neural network this paper proposes a solution to potato ailment classification through transfer learning using MobileNet architecture. A lot of research is done in the field of crop disease detection. Rice Blast Disease Recognition Using a Deep Convolutional Neural Network by Wan-jie Liang [2] details a study using 2906 positive and 2902 negative samples of rice for training and testing CNN model. The results obtained highlighted that CNN extracted in depth discriminative high-level features than LBPH (Local Binary Pattern Histogram Algorithm). The evaluation showed accuracy of 95.83% and 0.99% area under curve with CNN and CNN+SVM respectively over and LBPH+SVM with 82.59% accuracy and 0.9%area under curve. This revealed the importance of deep convolutional network over hand crafted features. Kahar [3] gained 74.21%accuracy using artificial neural network to identify the disease of rice plant. Identification of maize disease was done by Kai [4] using a kernel function based on radial in SVM. Three categories of disease were reported with 89.6%

accuracy with training and test sample divide on 262 images. Mohanty [5], Kawaski [6], Zhang [7] used CNN to achieve plant disease detection in their work. Mohanty et al identified plant disease of 26 types with 14 types of species having 99.35% accuracy. Kawaski et al. identified diseased cucumber leaves images with 94.9% accuracy. Zhang et al obtained 98.8% accuracy in the identification maize plant disease. Thus, deep convolutional network has provided a way to easily identify and classify plant disease. Architecture of MobileNet and its lightweight features are discussed in detail in the coming section which provides a promising answer for its use in real time applications. The implementation of transfer learning is then highlighted in the section 3, which is carried out by freezing the whole model except some top layers. The result and observation section discusses the significant results along with parameters loss, accuracy and F1 score for evaluating the model's performance on train and validation dataset.

2 MATERIAL AND METHODS

2.1 Dataset

The experimental dataset was taken from plant village which is publicly available. The dataset provided the required categories of images meeting the criteria of experiment.

2.2 Preprocessing

Image preprocessing was performed to clean the images and exclude the unnecessary information. Segmented images are used as input to the deep learning MobileNet model as image segmentation makes the image richer in useful information. The images of potato leaves available from plant village- diseased by early blight, late blight is 1000 in numbers respectively while the number of healthy images were 152. In order to address class imbalance in data, data augmentation on minority class was applied by rotating the images in different direction which increased number of healthy images to 1000. The augmentation contains different methods like shifting, scaling, flipping, random rotation, random brightness, random zoom and many more. To maintain the uniformity in dataset only rotation was chosen as augmentation method for healthy images to increase their number. Rotation angles were 45, 90, 135, 180, 225, 270 degrees. Some sample images from prepared dataset are shown in Figure 1. After the process of augmentation, the dataset is then split into train, validation, and test sets. Referring to flow diagram in Figure 2. These sets are then preprocessed through keras library. The simulation parameters are selected as batch size number 16 and 32 (for testing the model on different conditions), size of image as 224x224. The batch size number ensures that at every time when model takes random images from theses sets it will be only 16 in number or32 in number. Lastly, transfer learning on pretrained MobileNet model is performed so that an optimum model can be developed which can successfully classify the potato leaves.

2.3 Deep Neural Network- MobileNet

The tradition followed is to increase accuracy by making network dense and complicated however the improvement in accuracy is not necessarily related to making more efficient network with respect to speed and size [8, 9]. In numerous genuine applications, for example, mechanical technology, self-driving vehicle and augmented

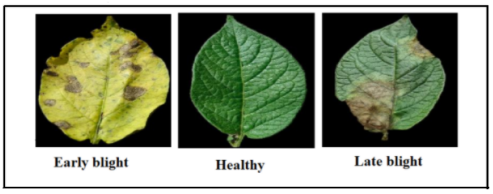


Figure 1: Sample photos from dataset

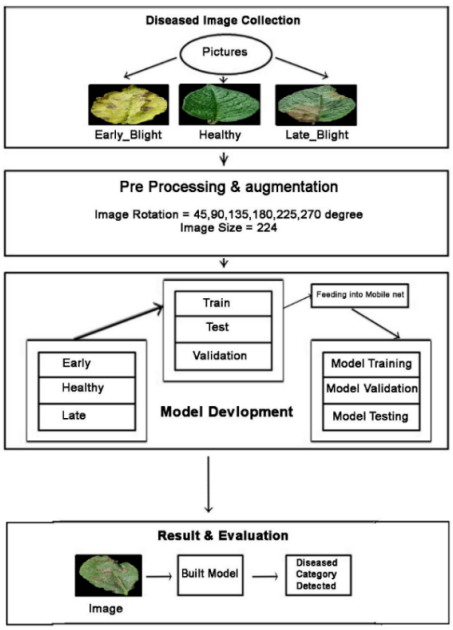


Figure 2: Overview of MobileNet based transfer learning system for Potato disease detection

reality, the recognition tasks should be done in an opportune manner on a computationally restricted device. MobileNet allows model developer to choose a smaller network that fits in the resource restriction for the application. These resource restrictions are latency and size. MobileNet's principally centered around improving latency yet, in addition it is lightweight network [10].

2.3.1 Architecture of MobileNet. This section explains about the composition of fundamental layers in MobileNet. These are designed using depth wise separable convolutions.

- a) Depth Wise Separable Convolutions

Unlike Spatial separable convolution that corresponds only to width and height depth wise separable convolution involves number of channels that include depth along with spatial dimension. Each channel represents particular elucidation of the given image. The three channels, RGB correspond to red, green, and blue component of each point in image respectively so an image with 256 channels has 256 particular elucidation of the image.

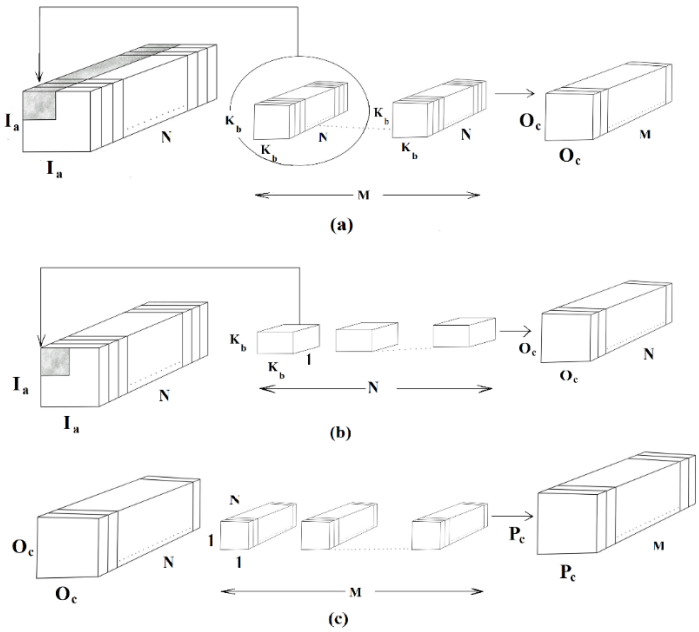


Figure 3: Depthwise separable convolutions in MobileNet architecture

Unlike the normal convolution, the development of depth wise separable convolution is based on factorization of normal convolution into two layers depth wise convolution and 1x1 convolution called point convolution.

Depth wise convolution in MobileNet applies one filter per input channel, 1x1 pointwise convolution combines the output from depth wise convolution. The number of channels of the final output image can be increased by increasing the depth of 1x1 pointwise convolution. This factorization has the effect of drastically reducing computation cost and size of model unlike the standard convolution where filters and inputs are combined to produce output in one step which has much higher computation cost.

- b) Comparison of Computational cost of Depth wise Separable Convolution with Standard convolution

Referring to diagram in Figure 3a. A standard convolutional layer takes input feature map I with I_a as spatial width and height, N as number of input channels of input feature map (I_a x I_a x N) and produces output feature map O with O_c as spatial width and height, M as number of output channels of output feature map (O_c x O_c x M). The parameter of standard convolutional is defined on convolutional kernel K as (K_b x K_b x M x N) where K_b as spatial width and height of convolutional kernel assumed to be square, N is the number of input channels and M as the number of output channels as defined earlier.

Referring to diagram in Figure 3a. Assuming stride as 1 and padding present the standard convolution has output feature map as computed by (1)

$$O_{c,j} = \left(\sum_{j=0}^{m} \left(\sum_{i=0}^{n} K_{b,i,j} * I_{a,i}\right)\right) \quad (1)$$

The computational cost is- (K_b.K_b.M.N.I_a.I_a) having dependency on multiple factors -size of kernel, number of output channels, number of input channels, size of input feature map.

The operation of standard convolution is to filter the input features using convolutional kernel and combining the filtered feature to depict new portrayal. This filtering and combining is done in single step.

The high computational cost of standard convolution can be reduced by using depthwise separable convolution which factorizes the standard convolution into depthwise convolution and point wise convolution.

Referring to diagram in Figure 3b. The post filtered channel of input image (1 channel per filter) is stacked together to form output image of depthwise convolution. Pointwise convolution, a 1x1 convolutional kernel uses this output image as an input to iterate through every point. The by default depth of 1x1 kernel is the number of channels present in input image. The number of kernels (with specified depth) in pointwise convolutional then can be increased to produce a linear combination of output image from depth wise layer.

The computation of depth wise convolutional with one filter per input channel can be shown by (2).

$$O_{c,i} = \sum_{i=0}^{n} \left(K_{b,i} * I_{a,i}\right) \quad (2)$$

where K_b is spatial width and height of depth wise convolutional filter, nth filter is applied to nth channel of input image, I_a is spatial width and height of input image, O_c is spatial width and height of output image.

It has computation cost as- K_b. K_b. N. I_a.I_a

Then, the output image from depth wise convolution is given as input to 1x1 pointwise convolution kernel K_1 as shown in Figure

3c and has computation given by (3).

$$P_{c,j} = \sum_{j=0}^{m} \left(\sum_{i=0}^{n} \left(O_{c,i} * K_{1,i,j} \right) \right) \quad (3)$$

where K_1 denotes that spatial width and height of kernel which is 1, O_c is spatial width and height of image given as input to pointwise convolution, n is the depth of 1x1 convolutional kernel, m is the number of convolutional kernel present and also the depth of final output image, P_c is the spatial width and height of final output image. It will have computation cost as- N.M. O_c. O_c

Hence total computation cost due to depthwise separable convolution is

$$K_b.K_b.N.I_a.I_a + N.M.O_c.O_c$$

which is the addition of depth-wise and pointwise computation cost and is much less than computation cost of standard convolution which is ($K_b.K_b.M.N.I_a$. I_a). Due to factorization in spatial dimension very little computation power is consumed in depthwise convolution. Hence, MobileNet is preferred in applications where lighter model is required.

As mentioned depthwise separable convolution is the basis of MobileNet. Each layer is followed by Batch Normalization and RELU nonlinearity layer except for the pooling, dropout and dense layer present at last. This network structure of MobileNet is used and trained for Potato leaves image feature extraction and classification.

2.4 Transfer Learning Method and Implementation

The images of leaves were placed in different folders depending on their category as early blight, late blight or healthy. The train sample contains a total of 2100 images belonging to these three categories. Validation and test sample have total of 300 and 600 images respectively. Since pre-trained MobileNet model is used to perform feature extraction and fine tuning, the model already contains the trained weights from ImageNet dataset [8] when loaded. The learned model is then used for feature extraction from new dataset which is potato leaves images in this case but the final classifier present on top of learned model depends on the classes present in dataset on which it was previously trained. In order to convert this model to work on potato leaves dataset the top 5 layers of pre trained model are removed and two dropout layers and final dense layer with softmax activation function having output as 3 belonging to the three classes of the dataset is added. After this structural change, the model is ready to be trained for feature extraction and classification on the dataset.

To propel the model to be specific to potato leaves dataset fine tuning is performed. The first few layers present in base convolutional network learn in extremely straight forward manner and these layers generally gather nonexclusive highlights from the input images while the top layers learn features specific to the dataset on which it is trained. Hence fine tuning allows the model to learn complex features which are specific to the new dataset provided.

Fine tuning process is done by freezing the whole model except the first 23 layers from top. Therefore these 23 layers retain fewer complex features from pretrained model but are trained further to improve performance. The remaining freezed layer weights do not update during training. Training of the final model in this manner enhances its performance on the dataset.

3 EXPERIMENTAL ANALYSIS, RESULTS AND DISCUSSION

The optimization algorithm, loss function, accuracy and learning rate used to improve model performance are discussed in this section.

3.1 Optimization Algorithm

By using the Adaptive Gradient Algorithm and Root Mean Square Propagation algorithm accuracy increased in few epochs but better results are obtained by adaptive moment estimation algorithm (Adam) which is the best substitute of stochastic gradient descent for training the deep learning model. Adam with learning rate as 0.0001 was applied to train the model.

3.2 Loss Function

Since the images present in dataset has to be classified in one of the three classes. It is multi-class classification task where model predicts the probability of each class and the class with highest probability is selected therefore "categorical cross entropy "was chosen as a loss function for the compilation of model. This loss function has minimum value when the predicted probability of the class is closer to the original class as given by (4).

$$L = -\sum_{i=1}^{i=3} p_i log p_i * \quad (4)$$

where L is the loss calculated, i represents the three classes on which dataset is divided, p_i is the probability of original three classes. The class present has probability 1 while rest have 0.
p^*_i is the probability predicted by model for the three classes respectively. Minus sign – indicates that loss gets minimized when the distribution gets closer.

3.3 Accuracy Function

For calculating the accuracy for the developed model "categorical accuracy" function is taken. This function uses one hot label. When the label of original class matches with the label of predicted class having the highest probability it is considered as a success and is counted. After completing this process for each and every image the total count is then divided by the total number of images and hence the accuracy is calculated.

3.4 Results and Observation

Table 1 shows the splitting of dataset in the ratio 7:1:2 for train, validation and test.

After compiling the model using Adam as optimization algorithm with learning rate 0.0001, loss function as categorical cross entropy and metrics as categorical accuracy. The model is then trained for 5 epochs and 10 epochs.

Table 2 shows the accuracy achieved by the MobileNet model in different conditions on train, validation and test sets.

Optimum accuracy of 99.83% was seen to be achieved on two sets of condition as shown in Table 2

The loss and accuracy curve at Learning rate=0.0001, batch size=32, epoch=10 is shown in Figure 4 which provided 99.83% accuracy on test dataset of 600 images.

Table 1: Dataset split

Class	Early Blight	Late Blight	Healthy
No. of training images	700	700	700
No. of validation images	100	100	100
No. of test images	200	200	200

Table 2: Best performance model accuracy results

Simulation parameters	Train accuracy	Validation accuracy	Test accuracy
Learning rate=0.0001, batch size=16, epoch=10	99.96%	99.33%	99.66%
Learning rate=0.0001, batch size=32, epoch=10	99.96%	99.00%	99.83%

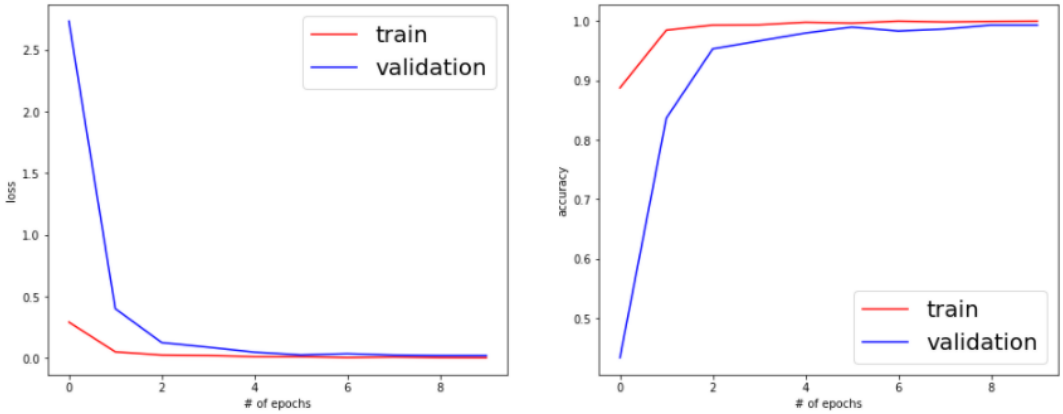


Figure 4: Loss and accuracy graph

Further in order to visualize performance of the classification model on test dataset, the confusion matrix showing the number of images classified correctly and incorrectly for each of the three classes was plotted. It is depicted in Figure 5. F1 Score as performance metric for the developed model is also calculated and its average value for three classes is .998, which is satisfactory.

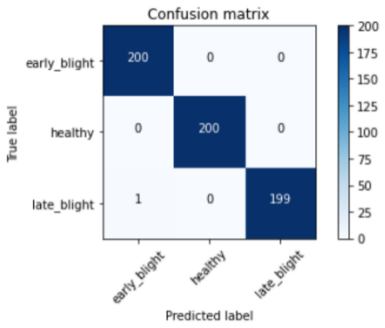


Figure 5: Confusion matrix

4 RELATED WORK

Significant prior work in the relevant area is listed in Table 3. D. Oppenheim et al. [11] developed a CNN model based on sliding window algorithm for classification of five classes with four diseased and one healthy class on a dataset of 2465 potato images.

Monzurul Islam et al. worked on an approach based on image segmentation followed by multiclass SVM and achieved accuracy is 95% on 300 images. Kurmi, Yashwant et al developed a multilayer perceptron model on plant village dataset and accuracy obtained is 91.22% with AUC of 92 %. Another work by Divyansh Tiwari et al. 2020 [14] used transfer learning-based models' inception V3, VGG16, VGG19 with different classifiers like KNN, SVM, Highest accuracy of 97.8% is achieved by using VGG19 with Logistic regression. This experiment conducted using MobileNet model obtained accuracy of 99.83%, which is highest and further the depth-wise separable convolutions in MobileNet result in lightweight deep neural network with reduced complexity that can be readily employed for real time applications.

Table 3: Performance comparison of proposed work with other published work

Related work	Method used	Performance
D. Oppenheim et al. 2017 [11]	CNN with sliding window algorithm neural network (BPNN) as a classifier	Accuracy 95.85%
Monzurul Islam et al. 2017 [12]	Image segmentation with multiclass SVM	96% in area under the ROC curve for classification
Kurmi, Yashwant et al. 2020 [13]	Multi-layer Perceptron	Accuracy 92%
Divyansh Tiwari et al. 2020 [14]	Models used inception V3, VGG16, VGG19 with all the different classifiers like KNN, SVM, Neural Network and Logistic regression	Highest accuracy achieved is 97.8%, by using VGG19 with Logistic regression
Proposed Work	MobileNet- V1 model with transfer learning	Accuracy 99.83%

5 CONCLUSION AND FUTURE WORK

In order to save time and money of small-scale farmers by keeping their agricultural production undisturbed it is absolute necessary that inexpensive, automatic, and easy to use disease detection system is built, which is capable of classifying the plant disease at an early stage, thereby informing farmer about the actual disease for the implementation of specific actions and treatment to increase the yield.

The above said objectives can only be achieved through technological means. The recent achievements in deep neural network have provided the path. MobileNet based on depth wise separable convolution is a pretrained model on a different large dataset and by using this Transfer learning approach, it was retrained on potato leaves dataset. Transfer learning approach was implemented by training top 23 layers while keeping the weight of bottom layers constant. The performed experiment showed 99.83% accuracy in the image classification task.

The future works involves successfully integrating it with portable devices, and possible inclusion of IoT framework to leverage cloud platforms for data storage and automation.

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